

Chapter 7

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Claim 7.1. *Let X and Y be topological spaces, and let $f : X \rightarrow Y$ be a function. Then the following are equivalent*

1. f is continuous function
2. For every closed set K in Y , $f^*(K)$ is closed in X
3. For every limit point p of a set A in X , $f(p) \in \overline{f_*(A)}$
4. For every $x \in X$ and a open set V containing $f(x)$, there is a open set U containing x such that $f_*(U) \subseteq V$

Proof. (1) $\implies$ (2): Suppose (1) is true. Let K be any closed set in Y . Then K^c is a open set. By definition of a continuous function, $f^*(K^c)$ is open. $\forall x \in f^*(K^c)$, $f(x) \notin K \implies x \notin f^*(K) \implies x \in X - f^*(K)$. Thus $f^*(K^c) \subseteq X - f^*(K)$. $\forall x \in X - f^*(K)$, $x \notin f^*(K) \implies f(x) \notin K \implies f(x) \in K^c \implies x \in f^*(K^c)$. Thus $X - f^*(K) \subseteq f^*(K^c) \implies X - f^*(K) = f^*(K^c) \implies X - f^*(K)$ is open. Thus $f^*(K)$ is closed.

(2) $\implies$ (3): Suppose (2) is true. $\forall A \subseteq X$, $\forall p \in X$, where p is a limit point of a A . $\overline{f_*(A)}$ is closed in Y , by 2 $f^*(\overline{f_*(A)})$ is closed in X . Since $A \subseteq \overline{f_*(A)}$, p is a limit point of $f^*(\overline{f_*(A)})$, but since $f^*(\overline{f_*(A)})$ is closed $p \in f^*(\overline{f_*(A)})$. Thus $f(p) \in \overline{f_*(A)}$.

(3) $\implies$ (1): Suppose (3) is true. The statement is the same as $\forall A \subseteq X$, $f(\overline{A}) \subseteq \overline{f(A)}$. $\forall U \in \mathcal{T}_Y$, $f^*(Y - U) = X - f^*(U)$. $f_*(\overline{X - f^*(U)}) \subseteq \overline{f_*(X - f^*(U))} = \overline{Y - U} = Y - U$. But since $X - f^*(U) \subseteq \overline{X - f^*(U)}$, and $\overline{f_*(X - f^*(U))} = Y - U \implies Y - U \subseteq \overline{f_*(X - f^*(U))} \implies f_*(\overline{X - f^*(U)}) = Y - U \implies \overline{X - f^*(U)} \subseteq X - f^*(U) \implies X - f^*(U) = \overline{X - f^*(U)}$. Thus $X - f^*(U)$ is closed and $f^*(U)$ is open.

Thus (1) $\implies$ (2) $\implies$ (3) $\implies$ (1).

(1) $\implies$ (4): Assume (1) is true. $\forall x \in X$ and $\forall V \in \mathcal{T}_Y \ni f(x) \in V$. Thus $U = f^*(V) \in \mathcal{T}_X$ by definition of continuous function. Thus $f(p) \in f_*(U) = V$.

(4) $\implies$ (1): Assume 4 is true. $\forall V \in \mathcal{T}_Y$, $\forall x \in f^*(V)$, $\exists U_x \in \mathcal{T}_X \ni x \in U_x \subseteq f^*(V)$. $\bigcup_{x \in f^*(V)} U_x = f^*(V)$ is open. Thus f is continuous.

Thus (1) $\implies$ (4) $\implies$ (1) □

Claim 7.2. *Let X, Y be topological spaces and $y_0 \in Y$. The constant map $f(x) = y_0$ for all $x \in X$*

Proof. $\forall U \in \mathcal{T}_Y$, if $y_0 \in U \implies X = f^*(U)$ else if $y_0 \notin U \implies U = \emptyset$ because $f_*(X) = \{y_0\}$. X and $\emptyset$ are respectively open in X . Thus f is continuous. □

Claim 7.3. Let $X \subseteq Y$ be topological spaces. The natural inclusion map is $i : X \rightarrow Y$ is continuous

Proof. By definition, for all subsets U of Y , $i^{-1}(U) = U \cap X$. Thus $\forall U \in \mathcal{T}_Y, i^{-1}(U) = U \cap X \in \mathcal{T}_X$ by definition of subspace topology. $\square$

Claim 7.5. A function $f : \mathbb{R}_n \rightarrow \mathbb{R}_n$ is continuous iff $\forall x \in \mathbb{R}$ and $\forall \epsilon > 0 \exists \delta > 0$ such that $\forall y \in \mathbb{R} \ni d(x, y) < \delta, d(f(x), f(y)) < \epsilon$.

Proof. $\implies$: Suppose $f : \mathbb{R}_n \rightarrow \mathbb{R}_n$ is continous. $\forall x \in \mathbb{R}$ and $\forall \epsilon > 0$, by defintion of standard topology on $\mathbb{R}$, $V = (f(x) - \epsilon, f(x) + \epsilon) \in \mathcal{T}_{\mathbb{R}_n}$. By Claim 7.1 1 $\implies$ 4, $\exists U \in \mathcal{T}_{\mathbb{R}_n} \ni x \in U$ and $f(U) \subseteq V$. By definition of basis, $\exists \delta \in \mathbb{R} \ni (x - \delta, x + \delta) \subseteq U \implies f_*((x - \delta, x + \delta)) \subseteq V$. Thus $\forall y \in \mathbb{R} \ni y \in (x - \delta, x + \delta) \implies f(y) \in V$. Thus the implication direction is true.

$\impliedby$: Suppose that a function $f : \mathbb{R}_n \rightarrow \mathbb{R}_n$ has the property that $\forall x \in \mathbb{R}$ and $\forall \epsilon > 0, \exists \delta > 0$ such that $\forall y \in \mathbb{R} \ni d(x, y) < \delta, d(f(x), f(y)) < \epsilon$. $\forall x \in \mathbb{R}_n, \forall V \in \mathcal{T}_{\mathbb{R}_n} \ni f(x) \in V$, by the definition of basis $V = \bigcup_{\alpha \in \lambda} (m_\alpha, n_\alpha)$. If $f(x) \notin (m_\alpha, n_\alpha)$, let $\epsilon_\alpha = \frac{|n_\alpha - m_\alpha|}{2}$ and let $p_\alpha = \frac{m_\alpha + n_\alpha}{2}$ thus $(m_\alpha, n_\alpha) \supset (p_\alpha - \epsilon_\alpha, p_\alpha + \epsilon_\alpha)$. If $f(x) \in (m_\alpha, n_\alpha)$, let $\epsilon_\alpha = \min(d(f(x), m_\alpha), d(f(x), n_\alpha))$ and let $p_\alpha = f(x)$ thus $(m_\alpha, n_\alpha) \supset (p_\alpha - \epsilon_\alpha, p_\alpha + \epsilon_\alpha)$. $\forall q \in f^*(\{p_\alpha\}), \exists \delta_q > 0 \ni \forall y \in \mathbb{R} \ni d(q, y) < \delta_q, d(f(y), p_\alpha) < \epsilon_\alpha$ thus $f_*((q - \delta_q, q + \delta_q)) \subseteq (p_\alpha - \epsilon_\alpha, p_\alpha + \epsilon_\alpha) \subseteq V$. Thus $U_\alpha = \bigcup_{q \in f^*(\{p_\alpha\})} (q - \delta_q, q + \delta_q)$ and $f(U_\alpha) \subseteq V$. Thus $U = \bigcup_{\alpha \in \lambda} U_\alpha$ and $f(U) \subseteq V$. Thus by 7.1, f is continuous. $\square$

Claim 7.6. Let X be a 1st countable space, and let Y be a topological space. Then $f : X \rightarrow Y$ is continuous iff for every convergent sequence $x_n \rightarrow x$ in X , $f(x_n) \rightarrow f(x)$ in Y .

Proof. $\implies$: Suppose f is continous. $\forall x \in X, \forall (x_n) \subseteq X$ such that $x_n \rightarrow x$. $\forall V \in \mathcal{T}_Y \ni f(x) \in V$, by the definition of continuity $\exists U \in \mathcal{T}_X \ni x \in U$ and $f_*(U) \subseteq V$. By the defintion of convergent sequences, $\exists K \in \mathbb{N} \ni \forall n \geq K, x_n \in U$ and since $f_*(U) \subseteq V \implies f(x_n) \in V$. Thus $f(x_n) \rightarrow f(x)$.

$\impliedby$: Suppose for every convergent sequence $x_n \rightarrow x$ in X , $f(x_n) \rightarrow f(x)$ in Y . $\forall A \subseteq X$, let $\forall x \in \delta A$, all limit points. By Theorem 5.18, $\exists (x_n) \subseteq A \ni x_n \rightarrow x$. By assumption $f(x_n) \rightarrow f(x)$. Note $(f(x_n)) \subseteq f_*(A)$. By Theorem 2.30, $f(x) \in f_*(A)$. Thus by Theorem 7.1 (Alternate definitions of continuity), f is continuous. $\square$

Claim 7.7. Let X be a space with a dense set D , and let Y be Hausdorff. Let $f : X \rightarrow Y$ and $g : X \rightarrow Y$ be continuous functions such that for every a in D , $f(a) = g(a)$. Then for all $x \in X$, $f(x) = g(x)$.

Proof. Assume the contrary that $\exists x \in X \ni f(x) \neq g(x)$. By the definition of Hausdorff, $\exists V_1, V_2 \in \mathcal{T}_Y \ni f(x) \in V_1, g(x) \in V_2, V_1 \cap V_2 = \emptyset$. By 7.1, $\exists U_1, U_2 \in \mathcal{T}_X \ni x \in U_1 \cap U_2$ and $f_*(U_1) \subseteq V_1$ and $f_*(U_2) \subseteq V_2$. By definition of dense set, $\exists d \in D \ni d \in U_1 \cap U_2$. $g(d) = f(d) \in V_1$ and $f(d) = g(d) \in V_2$ thus $f(d) = g(d) \in V_1 \cap V_2$. However this contradicts the fact that $V_1 \cap V_2 = \emptyset$. Thus a contradiction occurs, and thus $\forall x \in X, f(x) = g(x)$. $\square$

Claim 7.8. The cardinality of $\text{Hom}_{\text{cont}}(\mathbb{R}, \mathbb{R})$ is the same as $\mathbb{R}$.

Proof. By the previous theorem, all continous functions are completely determined by where they send each element of $\mathbb{Q}$ since $\mathbb{Q}$ is dense in $\mathbb{R}$ and $\mathbb{R}$ is Hausdorff. Thus for each $q \in \mathbb{Q}$ you are choosing out of $|\mathbb{R}| = 2^{\aleph_0}$ values. Thus there are at most $(2^{\aleph_0})^{\aleph_0} = 2^{\aleph_0 * \aleph_0} = 2^{\aleph_0} = |\mathbb{R}|$. There are $|\mathbb{R}|$ constant functions. $\square$

Claim 7.9. If $f : X \rightarrow Y$ and $g : Y \rightarrow Z$ are continuous, then their composition $g \circ f : X \rightarrow Z$ is continuous.

Proof. Let $W \in \mathcal{T}_Z$, by the continuity of g , $g^{-1}(W) \in \mathcal{T}_Y$. By continuity of f , $f^{-1}(g^{-1}(W)) \in \mathcal{T}_X$. $f^{-1}(g^{-1}(W)) = (f^{-1} \circ g^{-1})(W) = (g \circ f)^{-1}(W)$. Thus $g \circ f$ is continuous. $\square$

Claim 7.10. Let $X = A \cup B$, where A, B are closed in X . Let $f : A \rightarrow Y$ and $g : B \rightarrow Y$ be continuous functions that agree on $A \cap B$. Then the function $h : A \cup B \rightarrow Y$ defined by $h \equiv f$ on A and $h|_B \equiv g$ is continuous.

Proof. For every closed subset K of Y , $h^{-1}(K) = (h^{-1}(K) \cap A) \cup (h^{-1}(K) \cap B)$. $h^{-1}(K) \cap A = f^{-1}(K)$ and $h^{-1}(K) \cap B = g^{-1}(K)$ are both closed in A and B respectively by continuity of f and g . Since A and B are closed, $f^{-1}(K)$ and $g^{-1}(K)$ are closed. By finite union of closed sets is closed, thus $h^{-1}(K)$ is closed. Thus h is continuous. $\square$

Claim 7.11. Let $X = A \cup B$, where A, B are open in X . Let $f : A \rightarrow Y$ and $g : B \rightarrow Y$ be continuous functions that agree on $A \cap B$. Then the function $h : A \cup B \rightarrow Y$ defined by $h \equiv f$ on A and $h|_B \equiv g$ is continuous.

Proof. For every open subset K of Y , $h^{-1}(K) = (h^{-1}(K) \cap A) \cup (h^{-1}(K) \cap B)$. $h^{-1}(K) \cap A = f^{-1}(K)$ and $h^{-1}(K) \cap B = g^{-1}(K)$ are both open in A and B respectively by continuity of f and g . Since A and B are open, $f^{-1}(K)$ and $g^{-1}(K)$ are open. By finite union of open sets is open, thus $h^{-1}(K)$ is open. Thus h is continuous. $\square$

Claim 7.12. Let $f : X \rightarrow Y$ be a function and $\mathcal{B}$ be a basis for Y . Then f is continuous iff for every open set $B \in \mathcal{B}$, $f^{-1}(B) \in \mathcal{T}_X$.

Proof. $\implies$: f is continuous directly means that $\forall B \in \mathcal{B}, f^{-1}(B) \in \mathcal{T}_X$.
 $\impliedby$: Suppose $\forall B \in \mathcal{B}, f^{-1}(B) \in \mathcal{T}_X$. $\forall V \in \mathcal{T}_Y, V = \bigcup_{\alpha \in \lambda} B_\alpha$ for $B_\alpha \in \mathcal{B}$. $f^{-1}(V) = f^{-1}(\bigcup_{\alpha \in \lambda} B_\alpha) = \bigcup_{\alpha \in \lambda} f^{-1}(B_\alpha) \in \mathcal{T}_X$. Thus f is continuous. $\square$

Claim 7.13. Let $f : X \rightarrow Y$ be a function and $\mathcal{B}$ be a subbasis for Y . Then f is continuous iff for every open set $B \in \mathcal{B}$, $f^{-1}(B) \in \mathcal{T}_X$.

Proof. $\implies$: f is continuous directly means that $\forall B \in \mathcal{B}, f^{-1}(B) \in \mathcal{T}_X$.
 $\impliedby$: Suppose $\forall B \in \mathcal{B}, f^{-1}(B) \in \mathcal{T}_X$. For all basis elements V of $\mathcal{T}_Y$, $V = \bigcap_{i=1}^n B_i$ for $B_i \in \mathcal{B}$. $f^{-1}(V) = f^{-1}(\bigcap_{i=1}^n B_i) = \bigcap_{i=1}^n f^{-1}(B_i) \in \mathcal{T}_X$. By 7.12, f is continuous. $\square$

Properties Preserved by Continuous Functions

Claim 7.15. *If X is compact and $f : X \rightarrow Y$ is continuous and surjective, then Y is compact*

Proof. For any $\{V_\alpha\}_{\alpha \in \lambda}$ open cover, $\{f^{-1}(V_\alpha)\}_{\alpha \in \lambda}$ is a open cover of X because $f^{-1}(Y) = X$. By the compactness of X , there is a finite subcover $\{f^{-1}(V_{\alpha_i})\}_{i=1}^n$ of $\{f^{-1}(V_\alpha)\}_{\alpha \in \lambda}$. Since $X = \bigcup_{i=1}^n f^{-1}(V_{\alpha_i})$ and since f is surjective, $f(\bigcup_{i=1}^n f^{-1}(V_{\alpha_i})) = \bigcup_{i=1}^n f(f^{-1}(V_{\alpha_i})) = \bigcup_{i=1}^n V_{\alpha_i} = Y$. Thus there is a finite subcover of Y . $\square$

Claim 7.16. *If X is Lindelof and $f : X \rightarrow Y$ is continuous and surjective, then Y is Lindelof*

Proof. Pretty similar to 7.15 $\square$

Claim 7.17. *If X is Lindelof and $f : X \rightarrow Y$ is continuous and surjective, then Y is Lindelof*

Proof. Pretty similar to 7.15 and 7.16 $\square$

Claim 7.18. *Let D be a dense set of a topological space X , and let $f : X \rightarrow Y$ be continuous and surjective. Then $f(D)$ is dense in Y .*

Proof. $\forall V \in \mathcal{T}_Y \setminus \{\emptyset\}$, $f^{-1}(V)$ is open in X and is nonempty since f is surjective. By 5.1, $f^{-1}(V) \cap D \neq \emptyset$. Thus $\exists x \in f^{-1}(V) \cap D$. Thus $f(x) \in f(f^{-1}(V) \cap D) = V \cap f(D)$. Thus by 5.1, $f(D)$ is dense in Y . $\square$

Claim 7.21. *If X is normal and $f : X \rightarrow Y$ is continuous, surjective, and closed, then Y is normal.*

Proof. $\forall A, B \in \mathcal{C}_Y \ni A \cap B = \emptyset$. $f^{-1}(A), f^{-1}(B) \in \mathcal{C}_X$.

$$f^{-1}(A) \cap f^{-1}(B) = f^{-1}(A \cap B) = f^{-1}(\emptyset) = \emptyset$$

By Theorem 4.10, by the normality of X , $\exists U, V \in \mathcal{T}_X \ni A \subseteq U, B \subseteq V, \overline{U} \cap \overline{V} = \emptyset$. By closure of f , $f(\overline{U}), f(\overline{V}) \in \mathcal{C}_Y$. $f(\overline{U}) \cap f(\overline{V}) = f(\overline{U} \cap \overline{V}) = \emptyset$. Moreover by surjective, $A \subseteq f(f^{-1}(A)) \implies A \subseteq f(\overline{U})$ similarly $B \subseteq f(\overline{V})$.

By Theorem 4.10, Y is normal. $\square$

Claim 7.22. *If $\{B_\alpha\}_{\alpha \in \lambda}$ is a basis for X and $f : X \rightarrow Y$ is continuous, surjective, and open, then $\{f(B_\alpha)\}_{\alpha \in \lambda}$ is a basis for Y .*

Proof. By openness of f , $\{f(B_\alpha)\}_{\alpha \in \lambda} \subseteq \mathcal{T}_Y$. $\forall U \in \mathcal{T}_Y$ and $\forall p \in U$, by the surjectivity of f , $\exists x \in f^{-1}(p)$. By Theorem 3.1, $\exists B_\alpha \ni x \in B_\alpha \implies f(p) \in f(B_\alpha)$. By Theorem 3.1, $\{f(B_\alpha)\}_{\alpha \in \lambda}$ is a basis $\square$

Claim 7.23. *If X is 2nd countable and $f : X \rightarrow Y$ is continuous, surjective, and open, then Y is 2nd countable.*

Proof. By the 2nd countability of X , $\exists \{B_n\}_{n \in \mathbb{N}}$ a countable basis for X . By theorem 7.22, $\{f(B_n)\}_{n \in \mathbb{N}}$ is a basis for Y . Thus Y is 2nd countable. $\square$

Claim 7.24. *Let X be compact, and let Y be Hausdorff. Then any continuous function $f : X \rightarrow Y$ is closed.*

Proof. $\forall A \in \mathcal{C}_X$. $\forall x \in Y \setminus f(A)$. $\forall y \in f(A)$, by the Hausdorff property, $\exists U_y, V_y \ni x \in U_y, y \in V_y, U_y \cap V_y = \emptyset$. Thus there exists an open cover of $f(A)$, $\{V_y\}_{y \in f(A)}$ with a corresponding collection $\{U_y\}_{y \in f(A)}$. Thus $\{f^{-1}(V_y)\}_{y \in f(A)}$ is an open cover of A by continuity of f . By compactness of X , A is compact because it is closed. Thus there exists a finite subcover given by $y_1, \dots, y_n$ where $y_i \in f(A)$. Thus $\{f^{-1}(V_{y_i})\}_{i=1}^n$ is a finite subcover of A . Let $F = \bigcup_{i=1}^n f^{-1}(V_{y_i}) = f^{-1}(\bigcup_{i=1}^n V_{y_i}) \in \mathcal{T}_X$ and $G = \bigcap_{i=1}^n f^{-1}(U_{y_i}) = f^{-1}(\bigcap_{i=1}^n U_{y_i}) \in \mathcal{T}_X$ and $F \cap G = \emptyset$. Thus $f^{-1}(x) \not\subseteq \overline{A} = A$ and more specifically $G \cap A = \emptyset$. Thus $x \in f(G) = \bigcap_{i=1}^n f^{-1}(U_{y_i})$ and thus $f(G \cap A) = f(G) \cap f(A) = \emptyset$. Thus $x \notin \overline{f(A)}$. Thus $\overline{f(A)} = f(A)$ and $f(A)$ is closed. Thus f is a closed continuous map. $\square$

Claim 7.25. *Let X be compact and 2nd countable, and let Y be Hausdorff. If $f : X \rightarrow Y$ is continuous and surjective, then Y is 2nd countable.*

Proof. By claim 7.24, f is closed. Since Y is Hausdorff, singletons are closed. By continuity, inverse images of singletons are closed. Let $\mathcal{B}_X$ be the countable basis of X . WLOG, $\mathcal{B}_X$ is closed under finite union (countably many of these). $\forall B \in \mathcal{B}, V(B) := Y - f(X - B)$.

$\forall B \in \mathcal{B}, \forall b \in B, b \notin X - B \implies f(b) \notin f(X - B) \implies f(b) \in Y - f(X - B) \implies f(B) \subseteq Y - f(X - B)$

$\forall D \in \mathcal{T}_Y, \forall y \in D$. Since f is surjective, $f^{-1}(y) \neq \emptyset$. By continuity, $f^{-1}(D) \in \mathcal{T}_X$. For all $x \in f^{-1}(y)$, $\exists B_x \in \mathcal{B} \ni x \in B_x \subseteq f^{-1}(D)$. $\{B_x\}_{x \in f^{-1}(y)}$ is an open cover of $f^{-1}(y)$ which is compact because X is compact. Thus $\exists B_{x_1}, \dots, B_{x_n} \ni f^{-1}(y) \in B_{x_1} \cup \dots \cup B_{x_n} \subseteq f^{-1}(D)$. But since $\mathcal{B}$ is closed under finite unions, $\exists B \in \mathcal{B} \ni B = B_{x_1} \cup \dots \cup B_{x_n}$. Thus $f^{-1}(y) \subseteq B \subseteq f^{-1}(D)$. $\forall v \in V(B)$, by surjectivity $\exists u \in X, v = f(u)$. $f^{-1}(V(B)) = f^{-1}(Y - f(X - B)) = X - f^{-1}(f(X - B)) = X - (X - B) = B \implies u \in B \subseteq f^{-1}(D) \implies v \in D$. Thus $V(B) \subseteq D$. Thus $y \in V(B) \subseteq D$. By 3.1, $\{V(B_i)\}_{i=1}^n$ is a basis. By definition it is countable. Thus Y is 2nd countable. $\square$

Homeomorphisms

Claim 7.28. *If $f : X \rightarrow Y$ is continuous, the following are equivalent:*

1. f is a Homeomorphism
2. f is a closed bijection
3. f is a open bijection

Proof. 1 $\implies$ 2 : If f is a Homeomorphism, it is a bijection and its inverse, g , is continuous. Thus $\forall A \in \mathbb{C}_X, g^{-1}(A) \in \mathbb{C}_Y$ by continuity. Since f is a bijection, $f(A) = g^{-1}(A) \in \mathbb{C}_Y$. Thus f is a closed bijection.

2 $\implies$ 3 : If f is a closed bijection. $\forall A \in \mathcal{T}_X, f(X - A) \in \mathbb{C}_Y$. By surjectivity, $f(X - A) = Y - f(A) \in \mathbb{C}_Y \implies Y - f(X - A) = Y - (Y - f(A)) = f(A) \in \mathcal{T}_Y$. Thus f is a open bijection.

3 $\implies$ 1: If f is a open bijection. That means that there exists a inverse function g and that $\forall A \in \mathcal{T}_X \implies f(A) \in \mathcal{T}_Y$. Thus $\forall A \in \mathcal{T}_X$, since $g^{-1}(A) = f(A) \in \mathcal{T}_Y$. Thus g is continuous. Thus f is a continuous bijection with a continuous inverse function. Thus f is a Homeomorphism.

Thus 1 $\implies$ 2 $\implies$ 3 $\implies$ 1 □

Claim 7.29. *Suppose $f : X \rightarrow Y$ is a continuous bijection where X is compact and Y is Hausdorff. Then f is a homeomorphism.*

Proof. By 7.24, f is closed. Thus by 7.28, since f is a closed continuous bijection, f is a homeomorphism. □

Claim 7.31. *Let X be a compact space, and let Y be Hausdorff. If $f : X \rightarrow Y$ is a continuous, injective map, then f is an embedding.*

Proof. Let $Z = f(X) \subseteq Y$ and let Z have the subspace topology of Y . Thus redefine f as $f : X \rightarrow Z$, f is now a bijection. Thus f is still a continuous function. By 7.29, f is a homeomorphism. Thus $X \cong f(X) \subseteq Y$ naturally. □

Product Space and Continuity

Claim 7.32. *Let X and Y be topological spaces. The projection maps are continuous, surjective, and open*

Proof. It suffices to show that π_x is continuous, surjective, and open. $\forall U \in \mathcal{T}_X, \pi_X^{-1}(U) = U \times Y \in \mathcal{T}_{X \times Y}$. Thus π_X is continuous.

$\forall x \in X, \forall y \in Y, \pi_X(x, y) = x$. Thus π_x is surjective.

$\forall A \in \mathcal{T}_{X \times Y}, \exists \{U_\alpha\}_{\alpha \in \lambda} \subseteq \mathcal{T}_X, \{V_\alpha\}_{\alpha \in \lambda} \subseteq \mathcal{T}_Y \ni A = \bigcup_{\alpha \in \lambda} (U_\alpha \times V_\alpha)$.

Thus $\pi_X(A) = \pi_X(\bigcup_{\alpha \in \lambda} U_\alpha \times V_\alpha) = \bigcup_{\alpha \in \lambda} \pi_X(U_\alpha \times V_\alpha) = \bigcup_{\alpha \in \lambda} U_\alpha \in \mathcal{T}_X$. Thus π_X is open. $\square$

Claim 7.33. *Let X and Y be topological spaces. The product topology on $X \times Y$ is the coarsest topology on $X \times Y$ that makes the projection maps π_X, π_Y on $X \times Y$ continuous.*

Proof. Let $\mathcal{T} = \mathcal{T}_{X \times Y}$. Assume the contrary that $\exists \mathcal{T}'_{X \times Y}$ such that $\mathcal{T}' \subseteq \mathcal{T}$ such that π_X, π_Y is continuous on $X \times Y$. Thus $\exists A \in \mathcal{T} \setminus \mathcal{T}', A = \bigcup_{\alpha \in \lambda} (U_\alpha \times V_\alpha)$ where $U_\alpha \in \mathcal{T}_X$ and $V_\alpha \in \mathcal{T}_Y$. By continuity of $\pi_X, \pi_X^{-1}(U_\alpha) = U_\alpha \times Y \in \mathcal{T}'$. Similarly, $\pi_Y^{-1}(V_\alpha) = X \times V_\alpha \in \mathcal{T}'$. $A = \bigcup_{\alpha \in \lambda} \pi_X^{-1}(U_\alpha) \cap \pi_Y^{-1}(V_\alpha) \in \mathcal{T}'$ which is a contradiction. Thus $\mathcal{T}$ is the coarsest. $\square$

Claim 7.34. *Let X and Y be topological spaces. For every $y \in Y$, the subspace $X \times \{y\}$ of $X \times Y$ is homeomorphic to X .*

Proof. $f := \pi_X|_{X \times \{y\}}$ is continuous by 7.4. $g : X \rightarrow X \times y$ where $g(x) = (x, y)$. f is open. Thus f is a continuous open bijection thus f is a homeomorphism by 7.28. $\square$

Claim 7.44. *Let X, Y, Z be topological spaces. A function $g : Z \rightarrow X \times Y$ is continuous if and only if $\pi_X \circ g$ and $\pi_Y \circ g$ are both continuous.*

Proof. $\implies$: If g is continuous, by 7.9 and 7.32, $\pi_X \circ g$ and $\pi_Y \circ g$ are continuous.

$\Leftarrow$: If $\pi_X \circ g$ and $\pi_Y \circ g$ are continuous. $\forall A \in \mathcal{T}_{X \times Y}, A = \bigcup_{\alpha \in \lambda} (U_\alpha \times V_\alpha) = \bigcup_{\alpha \in \lambda} (\pi_X^{-1}(U_\alpha) \cap \pi_Y^{-1}(V_\alpha)) \implies g^{-1}(A) = \bigcup_{\alpha \in \lambda} g^{-1}(\pi_X^{-1}(U_\alpha) \cap \pi_Y^{-1}(V_\alpha)) = \bigcup_{\alpha \in \lambda} ((\pi_X \circ g)^{-1}(U_\alpha) \cap (\pi_Y \circ g)^{-1}(V_\alpha)) \in \mathcal{T}_Z$, by assumption. $\square$

Claim 7.32. *Let $\prod_{\alpha \in \lambda} X_\alpha$ be a product of topological spaces $\{X_\alpha\}_{\alpha \in \lambda}$. The projection maps are continuous, surjective, and open*

Proof. Basically same proof as finite case. $\square$

Quotient Spaces

Claim 7.47. *The quotient topology actually defines a topology.*

Proof. $f : X \rightarrow Y$ is a surjective map. $f^{-1}(Y) = X$, $f^{-1}(\emptyset) = \emptyset$.

$f^{-1}(\bigcup_{\alpha \in \lambda} U_\alpha) = \bigcup_{\alpha \in \lambda} f^{-1}(U_\alpha) \in \mathcal{T}_X \implies \bigcup_{\alpha \in \lambda} U_\alpha \in \mathcal{T}_Y$. Same for finite intersections □

Claim 7.48. *Let X be a topological space, let Y be a set, and let $f : X \rightarrow Y$ be a surjective map. The quotient topology on Y is the finest (largest) topology that makes f continuous.*

Proof. Let $\mathcal{T}$ denote the quotient topology on Y .

For all topologies, $\mathcal{T}'$ on Y such that f is continuous, $\forall U \in \mathcal{T}'$ by definition of continuity $f^{-1}(U) \in \mathcal{T}_X \implies U \in \mathcal{T}$. Thus $\mathcal{T}' \subseteq \mathcal{T}$. Thus $\mathcal{T}$ is the finest topology that makes f continuous. □

Claim 7.49. *Let X and Y be topological spaces. A surjective, continuous map $f : X \rightarrow Y$ that is an open map is a quotient map.*

Proof. $\forall U \in 2^Y \ni f^{-1}(U) \in \mathcal{T}_X \implies f(f^{-1}(U)) \in \mathcal{T}_Y$, by openness of f . But since f is surjective, $f(f^{-1}(U)) = U$. Thus Y has the quotient topology wrt to X and f . Thus f is a quotient map. □

Claim 7.50. *Let X and Y be topological spaces. A surjective, continuous map $f : X \rightarrow Y$ that is a closed map is a quotient map.*

Proof. $\forall U \in 2^Y \ni f^{-1}(U) \ni \mathcal{T}_X \implies f(X - f^{-1}(U)) \in \mathcal{C}_Y$, by closedness of f . By surjectivity of f , $f(X - f^{-1}(U)) = Y - f(f^{-1}(U)) = Y - U \in \mathcal{C}_Y \implies U \in \mathcal{T}_Y$. Thus Y has the quotient topology wrt to X and f . Thus f is a quotient map. □

Problem 7.51. *Show with examples that not all quotient maps are open maps, and not all quotient maps are closed maps*

Solution. 1. Let $[0, 1)$ have the subspace topology of $\mathbb{R}$. $q : [0, 1) \rightarrow S^1$ where $x \rightarrow x \pmod{1}$. $[0, \frac{1}{2})$ is open in $[0, 1]$, $q([0, \frac{1}{2})) = [0, \frac{1}{2}) \cup \{1\}$. But $q^{-1}([0, \frac{1}{2}) \cup \{1\}) = [0, \frac{1}{2}) \cup \{1\}$ is not open.

2. $\mathbb{R}^2 \rightarrow \mathbb{R}$, projection to x axis. Image of the set $xy = 1$ is $\mathbb{R} \setminus \{0\}$ which is not closed in $\mathbb{R}$ bc $\pi^{-1}(\mathbb{R} \setminus \{0\}) = \mathbb{R} \setminus 0 \times \mathbb{R}$ which is not closed in $\mathbb{R}$. ■

Claim 7.53. *Let $f : X \rightarrow Y$ be a quotient map. Then a map $g : Y \rightarrow Z$ is continuous if and only if $g \circ f$ is continuous.*

Proof. $\implies$: If g is continuous. $g \circ f$ is continuous by 6.9

$\impliedby$: If $g \circ f$ is continuous. By definition of continuity, $\forall U \in \mathcal{T}_Z$, $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U)) \in \mathcal{T}_X \implies g^{-1}(U) \in \mathcal{T}_Y$ since f is a quotient map. Thus g is continuous. □

Urysohn's Lemma

Claim 7.56. *Let A and B be disjoint closed sets in a normal space X . Then for each rational $r \in [0, 1]$, there exists an open set U_r such that $A \subseteq U_0$, $B \subseteq (X - U_1)$ and for $r < s$, $\overline{U_r} \subseteq U_s$.*

Proof. Since $P = [0, 1] \cap \mathbb{Q}$ is countable, thus $\exists f$ a bijection from $\mathbb{N} \rightarrow P$. Let f be such a bijection such that $f(1) = 1$ and $f(2) = 0$. Define $P_n := \{f(1), \dots, f(n)\}$. Let $U_1 = X - B$. By theorem 4.9, $\exists U_0 \in \mathcal{T}_X \ni A \subseteq U_0 \subseteq \overline{U_0} \subseteq U_1$.

Base Case: For P_2 , $\forall p, q \in P_2 \ni p < q \implies p = 0, q = 1$ we have already defined U_p and U_q such that $\overline{U_p} \subseteq U_q$.

Inductive Hypothesis: Assume that P_n holds the property such that $\forall p, q \in P_n \ni p < q \implies \overline{U_p} \subseteq U_q$.

Inductive Step: Let $r = f(n+1)$, thus $P_{n+1} = P_n \cup \{r\}$. This means for P_{n+1} to hold the property above, we need U_r such that for all

1. $p \in P_n \ni p < r$ such that $\overline{U_p} \subseteq U_r$
2. $\forall q \in P_n \ni q > r$ that $\overline{U_r} \subseteq U_q$.

Since f is a bijection and $f(1) = 1$ and $f(2) = 0$, $r \neq 1, 0$, thus r is not the max or min of P_n . Thus there exists values greater than it and values less than it. Since P_n is finite, there is a minimum value greater than it and a maximum value less than it.

Let $m = \max(p \in P_n \mid p < r)$ and let $n = \min(q \in P_n \mid q > r)$. By Theorem 4.9, $\exists U_r \in \mathcal{T}_X \ni \overline{U_m} \subseteq U_r \subseteq \overline{U_r} \subseteq U_n$. $\forall p \in P_n \ni p < r \implies p \leq m \implies \overline{U_p} \subseteq U_m \subseteq U_r$. Similarly, $\forall q \in P_n \ni q > r \implies q \geq n \implies \overline{U_r} \subseteq U_n \subseteq U_q$. Thus P_{n+1} respects the property.

Thus by induction the theorem is true. □

Claim 7.57. *(Urysohn's Lemma) A topological space X is normal if and only if for each pair of disjoint closed sets A and B in X , there exists a continuous function $f : X \rightarrow [0, 1]$ such that $A \in f^{-1}(0)$ and $B^{-1} \in f^{-1}(1)$.*

Proof. $\implies$: If X is normal. $\forall A, B \in \mathcal{C} \ni A \cap B = \emptyset$. By 7.56, $\forall r \in [0, 1] \cap \mathbb{Q}$, $\exists U_r \in \mathcal{T}$ such that $\forall s > r$ $\overline{U_r} \subseteq U_s$ and $A \subseteq U_0$ and $B \subseteq X - U_1$. Let $\{U_r\}_{r \in [0, 1] \cap \mathbb{Q}}$ be such a collection. Let define a function $f : X \rightarrow [0, 1]$ as follows:

1. If $x \in X - U_1$ then $f(x) = 1$
2. Otherwise, $f(x) = \inf \{r \in \mathbb{Q} \cap [0, 1] \mid x \in U_r\}$

Thus since $B \subseteq X - U_1 \implies f(B) = \{1\}$. Similarly since $A \subseteq U_0 \implies f(A) = 0$.

$\forall a \in \mathbb{Q} \cap [0, 1]$, $\forall x \in \overline{U_a}$, then $\forall b \in [0, 1] \cap \mathbb{Q} \ni b > a$, $x \in \overline{U_a} \subseteq U_b$, thus $(a, 1] \cap \mathbb{Q} \subseteq \{r \in \mathbb{Q} \cap [0, 1] \mid x \in U_r\}$. By definition of inf, $f(x) = \inf \{r \in \mathbb{Q} \cap [0, 1] \mid x \in U_r\} \leq a$.

$\forall a \in \mathbb{Q} \cap [0, 1]$, if $x \notin U_a$. By construction $x \notin U_b$ for all $b \leq a$. Thus $[0, a] \cap \mathbb{Q} \cap \{r \in \mathbb{Q} \cap [0, 1] \mid x \in U_r\} = \emptyset$. By definition of inf, $f(x) \geq a$.

Now lets check for continuity of f . The subbasis of $[0, 1]$ looks like

$$\{[0, a) \mid 0 < a \leq 1\} \cup \{(a, 1] \mid 0 \leq a < 1\}$$

$\forall a \in (0, 1]. \forall x \in f^{-1}([0, a)), 0 \leq f(x) < a$. By the density of the rationals in the reals, $\exists r \in \mathbb{Q} \ni 0 \leq f(x) < r < a$. By the contrapositive of the fact that if $x \notin U_r \implies f(x) \geq r$, since $f(x) < r \implies x \in U_r$. $f(U_r) \subseteq f(\overline{U_r}) \subseteq [0, r] \subseteq [0, a)$. Thus $x \in U_r \subseteq f^{-1}([0, a))$. By Theorem 2.3, $f^{-1}([0, a))$ is open.

$\forall a \in [0, 1]. \forall x \in f^{-1}((a, 1]), a < f(x) \leq 1$. By the density of the rationals in the reals, $\exists r \in \mathbb{Q} \ni a < r < f(x) \leq 1$. By the contrapositive of the fact that if $x \in \overline{U_r} \implies f(x) \leq a$, since $f(x) > r \implies x \notin \overline{U_r} \implies x \in X - \overline{U_r}$. $f(X - \overline{U_r}) \subseteq f(X - U_r) \subseteq [r, 1] \subseteq (a, 1]$. Thus $x \in X - \overline{U_r} \subseteq f^{-1}((a, 1])$.

Since the inverse images of the subbasis are open, f is continuous by 7.14.

$\Leftarrow$: If $\forall A, B \in \mathcal{C} \ni A \cap B = \emptyset, \exists f$ a continuous function such that $A \in f^{-1}(0)$ and $B \in f^{-1}(1)$. $[0, \frac{1}{2})$ and $(\frac{1}{2}, 1]$ are disjoint open sets. By continuity, $f^{-1}([0, \frac{1}{2}))$ and $f^{-1}((\frac{1}{2}, 1])$ are disjoint open sets. $A \subseteq f^{-1}(0) \subseteq f^{-1}([0, \frac{1}{2}))$ and $B \subseteq f^{-1}(1) \subseteq f^{-1}((\frac{1}{2}, 1])$. Thus X is normal. $\square$

Claim 7.58. *Let X be a normal space, and let A be a closed subset of X . Let $f : A \rightarrow [0, 1]$ be a continuous function, and let $r \in (0, 1)$. Then there exist disjoint open sets U_r and V_r such that $f^{-1}([0, r)) \subseteq U_r$ and $f^{-1}((r, 1]) \subseteq V_r$. Or equivalently, there exists an open set U_r such that $f^{-1}([0, r))$ and $\overline{U_r} \cap f^{-1}((r, 1]) = \emptyset$.*

Proof. $\forall r \in (0, 1)$, define $s_n = \max(r - \frac{1}{n}, \frac{r}{2})$ and $t_n = \min(r - \frac{1}{n}, \frac{r}{2} + \frac{1}{2})$. s_n and t_n are monotonically increasing and decreasing respectively and both converge to r in $[0, 1]$. Define, $S_n := f^{-1}([0, s_n))$ and $T_n := f^{-1}((t_n, 1])$.

$\forall x \in f^{-1}([0, r)) \implies 0 \leq f(x) < r$. Let $\epsilon = r - f(x)$, by definition of convergence $\exists K \in \mathbb{N} \ni \forall n \geq K, |r - s_n| = r - s_n < \epsilon$ (Monotonically increasing), thus $f(x) < s_n \leq r$. Thus $f(x) \in [0, s_n) \implies f(x) \in S_n$. Thus $x \in \bigcup_{n \in \mathbb{N}} S_n \implies f^{-1}([0, r)) \subseteq \bigcup_{n \in \mathbb{N}} S_n$.

$\forall x \in f^{-1}((r, 1]) \implies r < f(x) \leq 1$. By definition of convergence, $\exists K \in \mathbb{N} \ni \forall n \geq K, |r - t_n| = t_n - r < f(x) - r$, thus $r \leq t_n < f(x) < 1$. Thus $x \in f^{-1}((t_n, 1])$. Thus $x \in \bigcup_{n \in \mathbb{N}} T_n \implies f^{-1}((r, 1]) \subseteq \bigcup_{n \in \mathbb{N}} T_n$.

$\forall n \in \mathbb{N}, S_n \cap f^{-1}((r, 1]) = \emptyset$ since $[0, s_n) \cap (r, 1] = \emptyset$. For all limit points x of S_n . Assume the contrary $x \in f^{-1}((r, 1]) \implies 0 < r < f(x) \leq 1$. Thus $f^{-1}((f(x), 1]) \cap f^{-1}([0, s_n)) \neq \emptyset \implies [0, s_n) \cap (f(x), 1] \neq \emptyset$ even though $0 \leq s_n < r < f(x) \leq 1$ which is a contradiction. Thus by contradiction, $\overline{S_n} \cap f^{-1}((r, 1]) = \emptyset$.

By a symmetric argument, $\overline{T_n} \cap f^{-1}([0, r)) = \emptyset$

By the normality lemma, $\exists U_r, V_r \in \mathcal{T}$ such that the theorem is true. $\square$

Claim 7.59. *Tietze's Extension Theorem*

Proof. WLOG, target of f is $[-1, 1]$, simple transformation applied to $[0, 1]$. Let $I_{r,1} = [-r, -\frac{r}{3}]$, $I_{r,2} = [-\frac{r}{3}, \frac{r}{3}]$, and let $I_{r,3} = [\frac{r}{3}, r]$.

Base Case: $f^*(I_{1,1}), f^*(I_{1,3}) \subseteq \mathcal{C}_A \implies f^*(I_{1,1}), f^*(I_{1,3}) \subseteq \mathcal{C}_X \ni f^*(I_{1,1}) \cap f^*(I_{1,3}) = \emptyset$. By Urysohn's lemma, $\exists g_1 : X \rightarrow [-\frac{1}{3}, \frac{1}{3}] \ni g_1(f^*(I_{1,1})) = -\frac{1}{3}$ and $g_1(f^*(I_{1,3})) = \frac{1}{3}$ where g_1 is continuous.

Thus, $\forall x \in X$

$$|g_1(x)| \leq \frac{1}{3}$$

and $\forall x \in A$

$$|f(x) - g_1(x)| < \frac{2}{3}$$

Thus, $f(x) - g_1(x) : X \rightarrow [-\frac{2}{3}, \frac{2}{3}]$.

Inductive hypothesis: Assume that, $|f(x) - \sum_{i=1}^n g_n| < (\frac{2}{3})^n$ thus $f_n(x) := f(x) - \sum_{i=1}^n g_n : X \rightarrow [(-\frac{2}{3})^n, (\frac{2}{3})^n]$

Inductive Step: By continuity, $f_n^* \left(I_{(\frac{2}{3})^n, 1} \right), f_n^* \left(I_{(\frac{2}{3})^n, 3} \right) \in \mathcal{C}_A$, thus, $f_n^* \left(I_{(\frac{2}{3})^n, 1} \right), f_n^* \left(I_{(\frac{2}{3})^n, 3} \right) \in \mathcal{C}_X \ni f_n^* \left(I_{(\frac{2}{3})^n, 1} \right) \cap f_n^* \left(I_{(\frac{2}{3})^n, 3} \right) = \emptyset$.

By Urysohn's lemma, $\exists g_{n+1}$ a continous function from $X \rightarrow [-(\frac{2}{3})^n \frac{1}{3}, (\frac{2}{3})^n \frac{1}{3}]$ such that $g_{n+1} \left(f_n^* \left(I_{(\frac{2}{3})^n, 1} \right) \right) = -(\frac{2}{3})^n \frac{1}{3}$ and $g_{n+1} \left(f_n^* \left(I_{(\frac{2}{3})^n, 3} \right) \right) = (\frac{2}{3})^n \frac{1}{3}$.

Thus,

$$|g_{n+1}(x)| \leq \left(\frac{2}{3} \right)^n \frac{1}{3}$$

and $\forall x \in A$,

$$|f_{n+1}(x)| = \left| f(x) - \sum_{i=1}^{n+1} g_i(x) \right| \leq \left(\frac{2}{3} \right)^{n+1}$$

Define,

$$F(x) = \sum_{n=1}^{\infty} g(x)$$

$\forall x \in X$,

$$|F(x)| \leq \sum_{n=1}^{\infty} |g(x)| \leq \frac{1}{3} \sum_{n=1}^{\infty} \left(\frac{2}{3} \right)^n = \frac{1}{3} \frac{1}{1} = 1$$

Uniform convergence of $\sum_{i=1}^n g_n(x) \rightarrow F(x)$ comes from Weirstrass M test. Thus since $|f(x) - \sum_{i=1}^n g_n(x)| < (\frac{2}{3})^n$ and $\sum_{i=1}^n g_n(x) \rightarrow F(x)$ for all $a \in A$. Thus $F(a) = f(a)$. By completeness of $C(X, R)$ with respect to norm F is continuous.

$\Leftarrow$: Let A and B be a disjoint pair of closed sets of X . Let $f : A \cup B \rightarrow [0, 1]$ where $f(A) = 0$ and $f(B) = 1$. f is continous by pasting lemma. There exists a continous extension F of f onto the whole space. Thus by Urysohn's lemma, X is normal. $\square$

Claim 7.60. *A space X is normal if and only if for every closed set $A \subseteq X$ and continuous function $f : A \rightarrow (0, 1)$, there exists a continuous function $F : X \rightarrow (0, 1)$ such that $F(x) = f(x)$*

Proof. $\Rightarrow$: Without loss of generality suppose $f : A \rightarrow (-1, 1)$ as $(0, 1) \cong (-1, 1)$ under a simple homeomorphism. Let $g : A \rightarrow [-1, 1]$ where $a \rightarrow f(a)$. By Tietze's Theorem, $\exists G : X \rightarrow [-1, 1]$ where G is continous and $G|_A = g$. Let $B = G^*(\pm 1)$, $G(A) = (0, 1) \Rightarrow A \cap B = \emptyset$. Let $h : A \cup B \rightarrow [0, 1]$ where $h(A) = 1$ and $h(B) = 0$ which is continous by the pasting lemma. By Tietze's theorem, $\exists H : X \rightarrow [-1, 1]$ where $\forall a \in A \cup B$, $h(a) = H(a)$. Thus $F(x) = G(x)H(x)$. If $x \in A$, $|F(x)| = |G(x)H(x)| = |G(x)| < 1$. If $x \notin A$, $|F(x)| = |G(x)H(x)| < |G(x)| \leq 1$. Thus $F(x) \in (-1, 1)$. Thus $F : X \rightarrow (-1, 1)$.

$\Leftarrow$: Pretty immediate from Tietze's Theorem since $(0, 1) \subseteq [0, 1]$. $\square$

Claim 7.61. *A space X is normal if and only if for every closed set $A \subseteq X$ and continuous function $f : A \rightarrow [0, 1]$, there exists a continuous function $F : X \rightarrow [0, 1]$ such that $F(x) = f(x)$*

Proof. $\implies$: There is a $G : X \rightarrow [0, 1]$ such that for all $a \in A$, $G(a) = f(a)$. So $G(A) = [0, 1]$. Let $B = G^*(1)$. $A \cap B = \emptyset$. Let $h : A \cup B \rightarrow [0, 1]$ where $h(A) = 1$ and $h(B) = 0$. By Teitze's Theorme, $\exists H$ a continous extension of h . Let $F : G * H$. F is continous and $X \rightarrow [0, 1]$.

$\impliedby$: Pretty immediate from Tietze's Theorem since $[0, 1] \subseteq [0, 1]$. $\square$

Claim 7.62. *A space X is normal if and only if for every closed set $A \subseteq X$ and continuous function $f : A \rightarrow [0, 1] \times [0, 1]$, there exists a continuous function $F : X \rightarrow [0, 1] \times [0, 1]$ such that $F|_A = f$.*

Proof. $\implies$: By Teitze's theorem, $\exists F_1, F_2$ continous functions from $X \rightarrow [0, 1]$ where $F_i|_A = \pi_i \circ f$. Let $F : X \rightarrow [0, 1]^2$ where $x \rightarrow (F_1(x), F_2(x))$. $\forall a \in A$, $F(a) = (F_1(a), F_2(a)) = ((\pi_1 \circ f)(a), (\pi_2 \circ f)(a)) = f(a)$. $\pi_1 \circ F = F_1$ and $\pi_2 \circ F = F_2$. By Theorem 7.36, F is continous.

$\impliedby$: $\forall A \in \mathcal{C}$ and $\forall f : A \rightarrow [0, 1]$ where f is continous. Let $g : A \rightarrow [0, 1] \times [0, 1]$ where $a \rightarrow (f(x), f(x))$. $\exists G$ continous extension such that $G|_A = g$. Thus $\pi_1 \circ g = f$ and $F = \pi_1 \circ G$ are continous and agree on A . Thus by teitze's theorem, X is normal. $\square$

Claim 7.62. *A space X is normal if and only if for every closed set $A \subseteq X$ and continuous function $f : A \rightarrow \prod_{\alpha \in \lambda} [0, 1]^\alpha$, there exists a continuous function $F : X \rightarrow \text{prod}_{\alpha \in \lambda} [0, 1]^\alpha$ such that $F|_A = f$.*

Proof. Kinda identical to above $\square$

Claim 7.63. *Let X be a normal space, and let A be a closed subspace of X homeomorphic to $[0, 1]$ with the usual topology. Then there exists a continuous function $r : X \rightarrow A$ such that for every $r|_A = \text{id}_A$.*

Proof. Since $A \cong [0, 1] \implies \exists f, f^{-1}$ continous bijections from $A \rightarrow [0, 1]$ and $[0, 1] \rightarrow A$ respectively where $f \circ f^{-1} = \text{id}_{[0, 1]}$, $f^{-1} \circ f = \text{id}_A$. By Tietze's Theorem, $\exists F : X \rightarrow [0, 1]$ continous where $F|_A = f$. Let $r := f^{-1} \circ F$. Thus $\forall a \in A$, $r(a) = f^{-1}(F(a)) = f^{-1}(f(a)) = \text{id}_A(a) = a$. $\square$

Claim 7.63. *Let X be a normal space, and let A be a closed subspace of X homeomorphic to S^1 with the usual topology. Then there exists a continuous function $r : X \rightarrow A$ such that for every $r|_A = \text{id}_A$.*

Proof. $\square$

Claim 7.64a. *Suppose X is normal and A is F_σ . Then there is a function $f : X \rightarrow [0, 1]$ which only vanishes on A*

Proof.

$$A = \bigcap_{n \in \mathbb{N}} U_n$$

Without loss of generality, $U_n \subseteq U_{n+1}$.

By Urysohn's lemma, $\exists f_n$ continous where $f_n(A) = 0$ and $f_n(X - U_n) = 1$.

$$F(x) = \sum_{n \in \mathbb{N}} \frac{1}{2^n} f_n(x)$$

$\forall x \in X$,

$$\frac{1}{2^n} f_n(x) \leq \frac{1}{2^n}$$

Thus, $0 \leq F(x) \leq \sum_{n \in \mathbb{N}} \frac{1}{2} = 1$.

$$\begin{aligned} \left| \sum_{i=1}^k \frac{1}{2^n} f_i(x) - \sum_{i=1}^n \frac{1}{2^n} f_i(x) \right| &= \sum_{i=n+1}^k \frac{1}{2^i} f_i(x) \\ &\leq \sum_{i=n+1}^{\infty} \frac{1}{2^i} f_i(x) \\ &\leq \sum_{i=n+1}^{\infty} \frac{1}{2^i} \rightarrow 0 \end{aligned}$$

as $n \rightarrow \infty$. Thus $\sum_{i=1}^n \frac{1}{2^i} f_i(x) \rightarrow F$ uniformly thus F is continuous. $A \subseteq U_i$ thus $\forall x \in A, f_i(x) = 0$. If $x \notin A \implies x \in X - U_n$ for some n thus $f(x) \geq \frac{1}{2^n} f_n(x) = \frac{1}{2^n} > 0$. Thus F perfectly vanishes on A . $\square$

Claim 7.64. Suppose X is perfectly normal. Then for each pair of disjoint closed sets A and B in X , there exists a continuous function $f : X \rightarrow [0, 1]$ such that $A = f^{-1}(0)$ and $B = f^{-1}(1)$.

Proof. Let f_A be the function which perfectly vanishes on A . Let f_B be the function which perfectly vanishes on B .

Let $f : X \rightarrow [0, 1]$ where,

$$f(x) = \frac{f_A(x)}{f_A(x) + f_B(x)}$$

f is continuous by definition.

$$\begin{aligned} f(x) = 0 &\iff f_A(x) = 0 \iff x \in A \\ f(x) = 1 &\iff f_A(x) = f_A(x) + f_B(x) \iff f_B(x) = 0 \iff x \in B \end{aligned}$$

Thus f satisfies the requirement. $\square$

Claim 7.65. Perfectly normal implies completely normal

Proof. Let A and B be separated sets, thus $\overline{A} \cap B = \overline{B} \cap A = \emptyset$.

By claim 7.64a, there are functions, f, g which vanish perfectly on $\overline{A}$ and vanish perfectly on $\overline{B}$ respectively. $h : f - g : X \rightarrow [-1, 1]$. $\overline{B} \subseteq h^*((0, 1])$ and $\overline{A} \subseteq h^*([-1, 0))$. $h^*((0, 1]) \cap h^*([-1, 0)) = \emptyset$. Thus, X is completely normal. $\square$

Claim 7.66. *Let X be a normal, T_1 space. Then X is homeomorphic to a subspace of $\prod_{\alpha \in \lambda} [0, 1]_\alpha$ for some λ , where each factor is the unit interval with the standard topology.*

Proof.

□

Claim 7.70. *A space X is completely regular and T_1 if and only if X can be embedded in $\prod_{\alpha \in \lambda} [0, 1]_\alpha$ for some λ , where each factor is the unit interval with the standard topology.*

Claim 7.71. *Given a locally finite open cover $U_{\alpha \in \lambda}$ of a normal, T_1 space X , there is a collection of corresponding continuous functions $\varphi_\alpha : X \rightarrow [0, 1]$ such that (i) each φ_α is zero outside U_α , and (ii) the φ_α pointwise add to 1.*